CH2 Combinatorial Methods

## 2.2 Counting principle

### Theorem 2.1 (Counting Principle)

If the set E contains n elements and the set F contains m elements,

then, there are n*m ways in which we can choose

### Theorem 2.2 (Generalized Counting Principle)

If we have sets $E_1, E_2, E_3, ..., E_n$

then, there are $n_1 \times n_2 \times n_3 \times ... \times n_n$ ways to choice

### Example 2.2

In tossing 4 fair dice, P( at least one 3 among these 4 dice) = ?

$N(S) = 6 \times 6 \times 6 \times 6 = 6^4$

$N(A^C) = 5 * 5 * 5 * 5 = 5^4$

### Example 2.6 (Standard Birthday Problem)

P(at least two among n people have the same birthdays)=

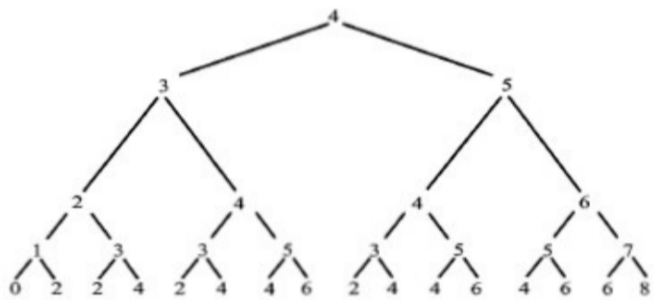
$N(S) = 365 * 365 * ... * 365 = 365^n$

$N(A^C) = 365 * 364 * ... * (365 - n + 1)$

### Counting principle: Tree Diagram

**Ex 2.9 Mark has $4. He decides to bet $1 on the flip of a fair coin 4 times. What is the probability that (a) he breaks even; (b) he wins money**

**Ans: (a) 6/16 (b) 5/16**



## 2.3 Permutations

An ordered arrangement of r object in set A containing n object is called:

r element permutation of A denote as ${}_nP_r,\ P(n,r),\ P_r^n$

$P_r^n = n \times (n-1) \times (n-2) \times \ldots \times (n-r)$

Ex 2.11: 2 electronics, 4 computer science, 3 statistics, 3 biology, and 5 music books are put on a bookshelf with a random arrangement. What is the probability the books of the same subject are together?

$\rightarrow$ 分組後在組內排序

$N(A) = 5! \times (2! \times 4! \times 3! \times 3! \times 5!)$

分成五個組別　每個組別內排序

Theorem 2.4

The number of distinguishable permutations of n objects with k different types

$$\frac{n!}{n_1! \times n_2! \times \ldots \times n_k!}$$

With replacement: 可以重複 (Sample space 不會變小)

Ex 2.15 A fair coin is flipped 10 times. P(exactly 3 heads) =

$N(S) = 210; N(A) = 10!/(3! * 7!)$

## 2.4 Combinations

### Definition

An unordered arrangement of r objects from a set of A containing n objects is called an r-element combination of A, denote as $_nC_r$, $C(n,r)$ $C^n_r$

### A Few Notes

$C^n_0 = C^n_n = 1$

$C^n_1 = C^n_{n-1} = n$

$C^n_r = C^n_{n-r}$ (10 個人，選3放室內與 選 7 放室外 是一樣的)

$C^{n+1}_r = C^n_r + C^n_{r-1}$ (設有 1 藍球與 9 白球，選 3 顆 的情況 $C^{10}_3$ 可以是)

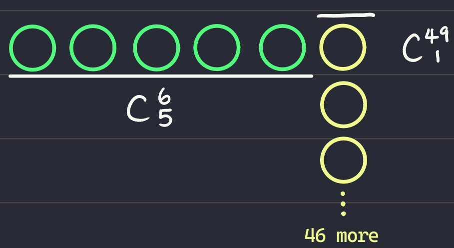
$\rightarrow$ 沒有包含藍球 $C^9_3$ 加上確定有藍球 $C^9_2$

Ex 2.17 45 instructors were selected randomly to ask whether they are happy with their teaching loads. The response of 32 were negative. If Drs. Smith, Brown, and Jones were among those questioned. P(all 3 gave negative responses)=?

Ex 2.19 In the lottery game, players pick 6 integers from 1 to 49, order of selection being irrelevant.

(a) P( Grand prize = 6 matches ) = ?

$P = 1 \div C^{49}_6$




(b) P( 2nd prize = 5 matches ) = ?

$P = (C^6_5 \times C^{43}_1) \div C^{49}_6$

Ex 2.20 7 cards are drawn from 52 without replacement.

P(at least one of the cards is a king)=?

算 $A^c$ ...

Ex 2.21 5 cards are drawn from 52. P(full house)=? $\boxed{A}$ $\boxed{A}$ $\boxed{A}$ $\boxed{B}$ $\boxed{B}$

Theorem (放隔板定理)

(a) The number of ways n indistinguishable objects can be placed into k distinguishable is

(b) The number of distinct nonnegative integer solutions that the equation $x_1 + x_2 + x_3 + ... + x_k = n$ has is

$$\binom{n+k-1}{n} = \binom{n+k-1}{k-1}$$

Theorem 2.5 (Binomial Theorem; Binomial expansion)(二項式定理)

$$(x+y)^n = \sum_{i=0}^{n} \boxed{\binom{n}{i}} x^{n-i} y^i$$

幾種該項的組合: $(x+y)^2 = x^2 + 2xy + y^2$

Example: What is the coefficient of the $x^2y^3$ in the expansion of $(2x+3y)^5$

$\binom{5}{3}$

Example: Evalutate the sum $\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n}$

將 二項式定理 x=1, y=1 代入 → $2^n$

Example: What is the coefficient of the $x^3y^2$ in the expansion of $(2x+3y)^5$

$$\sum_{i=0}^{n} (2x)^{n-i}(3y)^i$$

- **Ex 2.28 Prove that**

$$\binom{2n}{n} = \sum_{i=0}^{n} \binom{n}{i}^2 .$$

**Hint:**

$$2n = n + n; \quad \binom{n}{i} = \binom{n}{n-i}$$

Theorem 2.6 (Multinomial expansion)

$$(x_1 + x_2 + \cdots + x_k)^n = \sum_{n_1+n_2+\ldots+n_k=n} \frac{n!}{n_1!n_2!\ldots n_k!} x_1^{n_1} x_2^{n_2} \cdots x_k^{n_k}$$

This theorem is from Theorem 2.4